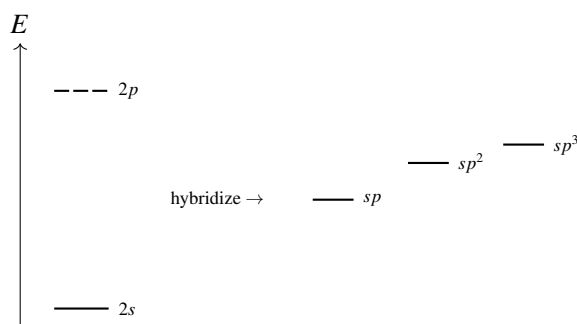
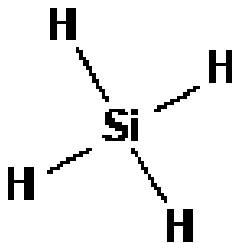


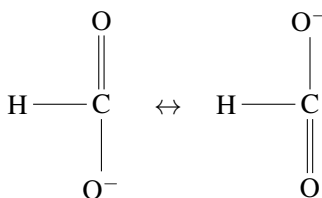
1. sp^3d^2 $\therefore$ the central atom has 6 electron groups, giving octahedral electron-group geometry with bond angles of approximately 90° .
2. The N–N in N_2H_4 is longer $\therefore$ it is a lower-order bond. N–N (146 pm, ~ 163 kJ/mol) vs. $N\equiv N$ (110 pm, ~ 945 kJ/mol); the triple bond has two additional π bonds, greatly increasing strength and decreasing length.
3. sp^3d^2 $\therefore$ Xe has 6 electron groups, requiring one s, three p, and two d orbitals to hybridize. $\therefore$ geometry: square planar, bond angles: 90° .
4. covalent $\therefore$ it consists only of nonmetals, whose atoms share electrons to form covalent bonds.
5. covalent network; high melting point $\therefore$ its atoms are linked in a continuous network of strong covalent bonds
6. A π bond forms by side-by-side overlap of p orbitals $\therefore$ lateral overlap is less effective than the head-on overlap of a σ bond, so less electron density lies between the nuclei.
7. No. $\therefore$ sp^3 hybridization mixes one s, 3 p orbital(s).
8. The sp^2 hybrid $\therefore$ it has the greater p -character, so its energy expectation value is weighted more toward the $2p$ orbitals; more s -character pulls a hybrid down toward the $2s$ energy.



9. Obeys the octet rule $\therefore$ the central Si atom is surrounded by 8 electrons

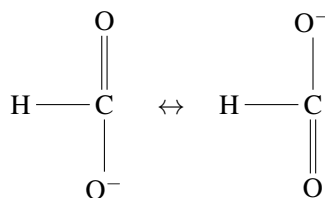


10. NOTE: this figure does not show any possible lone pairs and lone electrons. To get credit, YOU MUST show all existing lone pairs and lone electrons.
11. The claim is incorrect. Bond type depends on the orientation of orbital overlap, not on the orbital type alone. $\therefore$ two p orbitals overlapping end-to-end form a σ bond, whereas two p orbitals overlapping side-by-side form a π bond.
12. ionic $\therefore$ it consists of a metal and a nonmetal, so the particles are held together by ionic bonds.
13. Two occupied atomic orbitals (one on each atom) overlap, allowing two electrons of opposite spin to pair in the overlap region $\therefore$ the bond is a region of increased electron probability density between the nuclei, not a fixed pair of electrons.
14. 587 kJ $\therefore$ breaking bonds is endothermic; sum the bond energies of every reactant bond
15. sp hybridization $\therefore$ two Be–Cl single bonds; 2 electron groups, no lone pairs. $\therefore$ geometry: linear, bond angle: 180° .
16. False $\therefore$ it is a mathematical model of an atom's bonding orbitals, applied only once the atom is bonded — not a physical event.
17. No. $\therefore$ sp hybridization mixes one s, 1 p orbital(s).
- 18.



the bond order of each C–O bond is 1.5 $\therefore$ the bond order of a bond in a resonance hybrid is the average of its bond orders in the equivalent resonance structures; all of the resonance structures contribute equally, so every C–O bond is identical in the resonance hybrid. Sum the bond order of each C–O bond in HCO_2^- and divide by the number of equivalent resonance structures.

19. C: sp^2 — justified by 2 C–H bonds and 1 C=O bond
O: sp^2 — justified by 1 C=O bond and 2 lone pairs
20. NO is paramagnetic and weakly attracted $\therefore$ its 1 unpaired electron give a net magnetic moment that aligns with the field.
21. $\text{N}\equiv\text{N}$ (110 pm) < $\text{N}=\text{N}$ (122 pm) < $\text{N}-\text{N}$ (146 pm) $\therefore$ as bond order increases, additional shared electrons pull the nuclei closer together, decreasing bond length.
22. O_2 $\therefore$ adding electrons to O_2 fills the antibonding π^*_{2p} orbitals, reducing the bond order from 2 to 1 and weakening the bond.
23. $\text{KCl} < \text{NaCl} < \text{LiCl}$ $\therefore$ L.E. increases with greater ionic charge and smaller ionic radius (Coulomb's law $F \propto \frac{q_1 q_2}{r^2}$)
24. Cr, Fe, Ni, Cu $\therefore$ elements with greater metallicity hold their valence electrons less strongly because increased shielding and larger atomic radius reduce the nucleus-electron attraction. As a result, they have lower ionization energies and lose electrons more readily.
25. sp^3d $\therefore$ Cl has 5 electron groups, requiring one s , three p , and one d orbital to hybridize. $\therefore$ geometry: T-shaped, bond angles: $\leq 90^\circ$.
26. Answer: X belongs to Group 16 $\therefore$ X is S-like and has 6 valence electrons.
27. The following compounds have a dipole moment: NO_2
28. Yes $\therefore$ in C_2 the $2s$ and $2p$ orbitals are close enough in energy to mix, which pushes the π_{2p} orbitals below σ_{2p} .
- 29.



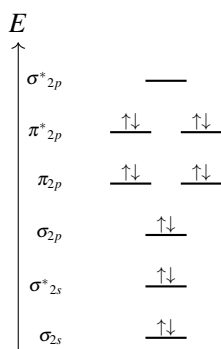
the bond order of each C–O bond is 1.5 $\therefore$ the bond order of a bond in a resonance hybrid is the average of its bond orders in the equivalent resonance structures; all of the resonance structures contribute equally, so every C–O bond is identical in the resonance hybrid. Sum the bond order of each C–O bond in HCO_2^- and divide by the number of equivalent resonance structures.

30. Approximately zero $\therefore$ at very large separation the atoms are essentially noninteracting, and this state is defined as the zero-potential-energy reference.
31. Bond order = 2 $\therefore$ there are 4 more bonding than antibonding electrons.
32. nonpolar covalent ($\Delta\text{EN} = 0.03$) $\therefore$ $\Delta\text{EN} < 0.4$, so the electrons are shared nearly equally and neither atom bears a partial charge
33. 8 valence electrons $\therefore$ sum the group valence electrons of every atom (subtract 1 for the overall charge); 0 lone pairs on the central N atom
34. 6 (one per carbon), all perpendicular to the ring $\therefore$ they overlap continuously to form a single delocalized π system above and below the plane of the ring.
35. 9 σ bonds and 1 π bond. The π bond comes from the C=O bond. $\therefore$ Counting the σ bond in every covalent bond and the additional π bond(s) present only in the multiple bond(s) gives these totals.

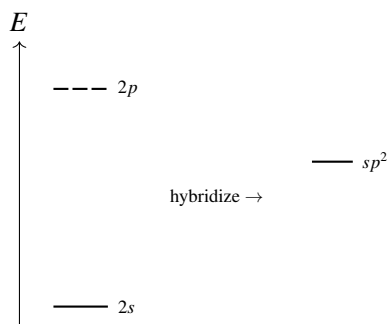


- 36.
37. sp^3d^2 $\therefore$ Xe has 6 electron groups, requiring one s , three p , and two d orbitals to hybridize. $\therefore$ geometry: square planar, bond angles: 90° .
38. +496 kJ $\therefore$ Hess's law: $\Delta H_f^\circ = \Delta H_{\text{sub}} + IE + \Delta H_{\text{diss}} + EA + \Delta H_{\text{lat}}^\circ$
39. MO theory $\therefore$ its MO bond order is 0 because bonding and antibonding electrons cancel.
40. -923 kJ $\therefore$ Hess's law: $\Delta H_f^\circ = \Delta H_{\text{sub}} + IE + \Delta H_{\text{diss}} + EA + \Delta H_{\text{lat}}^\circ$
41. $\text{C}\equiv\text{N} < \text{C}=\text{N} < \text{C}-\text{N}$ $\therefore$ higher bond order means more shared electron pairs between the nuclei, increasing the attraction between the atoms. Stronger attraction pulls the atoms closer together, so bond length decreases as bond order increases.
42. Answer: X belongs to Group 17 $\therefore$ X is Br-like and has 7 valence electrons.

43. Hybridization: sp^3 . Electron-Group Geometry: tetrahedral. Molecular Geometry: tetrahedral.
 $\therefore$ there are 4 bonds and 0 lone pairs around the central atom, there are 4 total electron groups surrounding it and $\therefore$ each requires one hybrid orbital, the central atom is sp^3 hybridized. The electron-group geometry is tetrahedral. Because all electron groups are bonding pairs, the molecular geometry is also tetrahedral with bond angles of 109.5° .
44. e^- -group arrangement: Trigonal planar
 Shape: Bent
 Ideal bond angle(s): $<120^\circ$
45. 26 valence electrons $\therefore$ sum the group valence electrons of every atom (add 1 for the overall charge); 1 lone pair on the central Cl atom
46. sp hybridization $\therefore$ two C=S double bonds give 2 electron groups around C. $\therefore$ geometry: linear, bond angle: 180° .
47. F_2 (BO=1) < O_2 (BO=2) < N_2 (BO=3). Individual bond orders: N_2 : BO=3, O_2 : BO=2, F_2 : BO=1. $\therefore$ bond order depends on the balance between electrons in bonding and antibonding molecular orbitals: electrons added to bonding orbitals increase bond order, whereas electrons added to antibonding orbitals decrease it. A complete justification correctly accounts for the MO occupancy of each species and uses it to explain the ranking, not merely state the bond-order values.
48. Bond order = $\frac{8-6}{2} = 1$; diamagnetic (0 unpaired electrons). The molecular orbitals are ordered using the correct mixing model and populated according to the Aufbau principle, the Pauli exclusion principle, and Hund's rule. The bond order follows from $\frac{N_b - N_a}{2}$, and the magnetic behavior follows from the presence or absence of unpaired electrons in the completed molecular-orbital diagram.



49. Head-on (end-to-end) overlap gives a σ bond, while side-by-side (lateral) overlap gives a π bond $\therefore$ the type of bond is set by how the orbitals overlap, not merely by which atoms are connected.
50. All have a bond order of 1. $\therefore$ they are isoelectronic, they occupy the same molecular orbitals with the same electron configuration, producing the same numbers of bonding and antibonding electrons and therefore the same bond order.
51. No $\therefore$ a σ bond forms from head-on overlap and provides the primary connection between the nuclei. It is energetically most favorable to form, so it will form before a π bond forms. A π bond forms from side-by-side overlap of parallel orbitals and can exist only in addition to an existing σ bond.
52. $+0 \therefore FC = \text{valence } e^- - \text{unshared } e^- - \frac{1}{2} \text{shared } e^- \therefore FC = 1 - 0 - \frac{1}{2}(2)$
53. The C–C single bond is longer $\therefore$ it is a lower-order bond. C–C (~154 pm, ~347 kJ/mol) vs. $C \equiv C$ (~120 pm, ~839 kJ/mol); three shared electron pairs in a triple bond attract both nuclei more strongly than one pair.
54. ionic ($\Delta EN = 3.0$); F carries the δ^- $\therefore \Delta EN \geq 2.0$ and F is the more electronegative atom
55. MO config: $\sigma_{1s}^2 \sigma_{1s}^{*2} \sigma_{2s}^2 \sigma_{2s}^{*2} \sigma_{2p}^2 \pi_{2p}^2 \pi_{2p}^2 \pi_{2p}^{*2} \pi_{2p}^{*1}$
 Bond order = $\frac{10-5}{2} = 2.5$
 F_2^+ is paramagnetic (1 unpaired e^-). The bond order is 2.5 because there are 10 bonding and 5 antibonding electrons. It is paramagnetic because one electron remains unpaired.
56. B_2^- : $\sigma_{1s}^2 \sigma_{1s}^{*2} \sigma_{2s}^2 \sigma_{2s}^{*2} \pi_{2p_x}^2 \pi_{2p_y}^1$; unpaired MO(s): a π_{2p} orbital; paramagnetic.
57. $C \equiv C < C = C < C - C$ $\therefore$ higher bond order means more shared electron pairs between the nuclei, increasing the attraction between the atoms. Stronger attraction pulls the atoms closer together, so bond length decreases as bond order increases.
58. sp^3d $\therefore$ S has 5 electron groups, requiring one s , three p , and one d orbital to hybridize. $\therefore$ geometry: seesaw, bond angles: $<90^\circ$ and $<120^\circ$.
59. $\langle E \rangle = \frac{1}{3}(-24) + \frac{2}{3}(-9) = -14 \text{ eV}$ $\therefore$ the expectation value weights each atomic-orbital energy by the orbital's fractional character; it lies between the $2s$ and $2p$ energies, nearer $2p$.

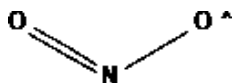


60.

61. Carbon: sp^2 — justified by 1 C–H bond, 1 C=O bond, and 1 C–O bond

Carbonyl oxygen: sp^2 — justified by 1 C=O bond and 2 lone pairs

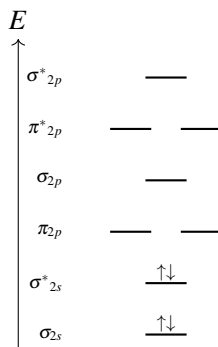
Hydroxyl oxygen: sp^3 — justified by 2 bonds and 2 lone pairs



62. NOTE: this figure does not show any possible lone pairs and lone electrons. To get credit, YOU MUST show all existing lone pairs and lone electrons.

63. (a) Electrons in a bonding MO stabilize the molecule (lower its energy); (b) electrons in an antibonding MO destabilize it (raise its energy) $\therefore$ bonding MOs build up density between the nuclei while antibonding MOs remove it.

64. Bond order = $\frac{2-2}{2} = 0$; diamagnetic (0 unpaired electrons). The molecular orbitals are ordered using the correct mixing model and populated according to the Aufbau principle, the Pauli exclusion principle, and Hund's rule. The bond order follows from $\frac{N_b - N_a}{2}$, and the magnetic behavior follows from the presence or absence of unpaired electrons in the completed molecular-orbital diagram. Bond order 0 predicts the molecule is not stable.



65. K_2Se $\therefore$ larger ion(s) $\rightarrow$ weaker electrostatic attraction; see Coulomb's law ($F \propto \frac{q_1 q_2}{r^2}$)

66. $1s^2 2s^2 2p^6 3s^2 3p^1$; Al forms Al^{3+} and is isoelectronic with Ne $\therefore$ given the neutral Al atom's electron configuration, it loses 3 electrons to match the noble-gas electron configuration of Ne.

67. 43.0% $\therefore$ percent ionic character = $\frac{\mu_{meas}}{\mu_{ionic}} \times 100$, where $\mu_{ionic} = e \cdot d = 5.14 \text{ D} \therefore \frac{2.21}{5.14} \times 100$

68. Hybridization. The hybridization determines the set of orbitals used for bonding and therefore the geometry around the atom. σ/π bonding describes the type of orbital overlap, not the spatial arrangement of the bonds.

69. $H_2 < H_2^- < H_2^+$. H_2 : σ_{1s}^2 ; unpaired MO(s): none; diamagnetic. H_2^- : $\sigma_{1s}^2 \sigma_{1s}^{*1}$; unpaired MO(s): σ_{1s}^* ; paramagnetic. H_2^+ : σ_{1s}^1 ; unpaired MO(s): σ_{1s} ; paramagnetic. Species with more unpaired electrons are more strongly attracted to a magnetic field.

70. MO config: $\sigma_{1s}^2 \sigma_{1s}^{*2} \sigma_{2s}^2 \sigma_{2s}^{*2} \sigma_{2p}^2 \pi_{2p}^2 \pi_{2p}^{*1} \pi_{2p}^{*1}$

Bond order = $\frac{10-6}{2} = 2$

O_2 is paramagnetic.

71. sp^3 $\therefore$ 4 electron groups require 4 hybrid orbitals (1 s + 3 p). $\therefore$ geometry: tetrahedral, bond angles: 109.5° .

72. Ti, Ga, Ge, As $\therefore$ elements with greater metallicity hold their valence electrons less strongly because increased shielding and larger atomic radius reduce the nucleus-electron attraction. As a result, they have lower ionization energies and lose electrons more readily.
73. N_2 $\therefore$ bond order $3 > 2$ for O_2 ; higher bond order means stronger, shorter, and more stable bonding.
74. $-93 \text{ kJ (exothermic)}$ $\therefore \Delta H^\circ \approx \sum BE_{\text{broken}} - \sum BE_{\text{formed}} = 2253 - 2346$
75. 3 $\therefore$ one antibonding σ^* orbital forms from end-on overlap and two antibonding π^* orbitals form from side-by-side overlap.
76. Cs $\therefore$ each step down a group adds an occupied electron shell
77. F_2 ($BO=1$) $< C_2$ ($BO=2$) $< N_2$ ($BO=3$). Individual bond orders: C_2 : $BO=2$, N_2 : $BO=3$, F_2 : $BO=1$. $\therefore$ bond order depends on the balance between electrons in bonding and antibonding molecular orbitals: electrons added to bonding orbitals increase bond order, whereas electrons added to antibonding orbitals decrease it. A complete justification correctly accounts for the MO occupancy of each species and uses it to explain the ranking, not merely state the bond-order values.
78. Bond order $= 0.5$ $\therefore$ there are 1.0 more bonding than antibonding electrons.
79. They represent the phase (sign) of the wavefunction $\therefore$ they do NOT represent positive or negative electric charge.
80. MO config: $\sigma_{1s}^2 \sigma_{1s}^{*2} \sigma_{2s}^2 \sigma_{2s}^{*2} \pi_{2p}^1 \pi_{2p}^1$
Bond order $= \frac{6-4}{2} = 1$
 B_2 is paramagnetic.
81. Obeys the octet rule $\therefore$ the central C atom is surrounded by 8 electrons
82. $KF < NaF < LiF$ $\therefore$ L.E. increases with greater ionic charge and smaller ionic radius (Coulomb's law $F \propto \frac{q_1 q_2}{r^2}$)
83. 36.9% $\therefore$ percent ionic character $= \frac{\mu_{\text{meas}}}{\mu_{\text{ionic}}} \times 100$, where $\mu_{\text{ionic}} = e \cdot d = 6.15 \text{ D}$ $\therefore \frac{2.27}{6.15} \times 100$
84. sp^2 $\therefore$ two C-H + one C=O; 3 electron groups around C. $\therefore$ geometry: trigonal planar, bond angles: $\approx 120^\circ$.
85. False $\therefore$ hybridization is applied to atoms once they are bonded; an isolated atom is described by its pure atomic orbitals.
86. MO theory $\therefore$ MO theory treats electrons as delocalized over all the nuclei.
87. sp^2 for the carbonyl C; sp^3 for the two methyl carbons. The carbonyl C has 3 electron groups (2 C-C + 1 C=O), giving sp^2 . Each methyl C has 4 single bonds, giving sp^3 $\therefore$ electron groups determine hybridization.
88. H_2^+ : MO configuration σ_{1s}^1 ; bond order $= \frac{1-0}{2} = 0.5$; paramagnetic.
 H_2 : MO configuration σ_{1s}^2 ; bond order $= \frac{2-0}{2} = 1$; diamagnetic.
Removing one electron changes the electron occupancy of the highest-energy MO. Adding an electron to an antibonding orbital *destabilizes* the molecule, *lowering* the bond order, while removing one from an antibonding orbital *stabilizes* the molecule, raising the bond order. An unpaired electron makes the species paramagnetic; if all electrons are paired, it is diamagnetic.
89. MO config: $\sigma_{1s}^2 \sigma_{1s}^{*2} \sigma_{2s}^2 \sigma_{2s}^{*2} \pi_{2p}^1$
Bond order $= \frac{5-2}{2} = 1.5$
 B_2^+ is paramagnetic (1 unpaired e^-). The bond order is 1.5 because there are 5 bonding and 2 antibonding electrons. It is paramagnetic because one electron remains unpaired.
90. The experimentally observed bond angle(s) are inconsistent with the orientations of unmodified atomic orbitals $\therefore$ hybridization mixes the S atom's s and p orbitals to produce sp^2 hybrid orbitals whose orientations are consistent with the observed geometry.
91. Only the placement of the π electrons (and the associated formal charges) changes between resonance structures. $\therefore$ the σ bonds and lone pairs around the central S remain the same, the electron-group count stays at 3. Because hybridization depends on those electron groups rather than on the location of the π bond, the central atom remains sp^2 .
92. Li_2^+ : MO configuration $\sigma_{1s}^2 \sigma_{1s}^{*2} \sigma_{2s}^1$; bond order $= \frac{3-2}{2} = 0.5$; paramagnetic.
 Li_2 : MO configuration $\sigma_{1s}^2 \sigma_{1s}^{*2} \sigma_{2s}^2$; bond order $= \frac{4-2}{2} = 1$; diamagnetic.
Removing one electron changes the electron occupancy of the highest-energy MO. Adding an electron to an antibonding orbital *destabilizes* the molecule, *lowering* the bond order, while removing one from an antibonding orbital *stabilizes* the molecule, raising the bond order. An unpaired electron makes the species paramagnetic; if all electrons are paired, it is diamagnetic.
93. Cis and trans isomers are distinct molecules $\therefore$ the rigid cis C=C bonds put kinks in unsaturated-fat chains, so they pack and behave differently from the flexible, all-single-bond chains of saturated fats.
94. A σ bond $\therefore$ head-on (end-to-end) overlap is more effective than side-by-side overlap, concentrating more electron density directly between the nuclei.
95. The $2s-2p$ energy gap is larger in O, F, and Ne $\therefore$ that gap is too wide for the σ_{2s} and σ_{2p} orbitals to mix; in Li-N the gap is small enough that mixing occurs, pushing the π_{2p} orbitals below σ_{2p} .

96. e^- -group arrangement: Octahedral
Shape: Octahedral
Ideal bond angle(s): 90°
97. e^- -group arrangement: Octahedral
Shape: Octahedral
Ideal bond angle(s): 90°
98. Approximately zero $\therefore$ at very large separation the atoms are essentially noninteracting, and this state is defined as the zero-potential-energy reference.
99. σ/π bonding. Rotational freedom depends on the type of orbital overlap present in the bond. A π bond requires side-by-side overlap that is lost upon rotation, whereas a σ bond retains end-to-end overlap.
100. e^- -group arrangement: Octahedral
Shape: Square planar
Ideal bond angle(s): 90°